

[Is it helpful for stress, anxiety, or sleep?: Fact Sheet for Health Professionals \(original version\)](#)

# Is it helpful for stress, anxiety, or sleep?: Fact Sheet for Health Professionals

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## Introduction

*Withania somnifera* (L.) Dunal is an evergreen shrub cultivated in tropical and subtropical areas of Asia, Africa, and Europe. It is commonly called by the Sanskrit name, ashwagandha, because the plant's roots are said to smell like a wet horse ("ashwa" for horse and "gandha" for smell) [1]. Ashwagandha is also sometimes known as winter cherry or Indian ginseng, although it does not belong to the ginseng family. Ashwagandha root has been used in the traditional Ayurvedic and Unani medicine systems of India as an adaptogen, which is loosely defined as a compound or product that increases the ability of a person to resist, adapt, or become resilient in nonspecific ways to biological, physical, or chemical stressors [2,3]. The chemical composition of ashwagandha root and leaf differ [4]. Most commercial ashwagandha supplements contain extracts from the plant's root, although some contain extracts from both the root and leaf.

The species name *somnifera* comes from the Latin word for sleep-inducing, signifying another purported property of this botanical [5]. In addition to sleep, ashwagandha is commonly promoted for stress and anxiety reduction.

Ashwagandha is rich in phytochemicals, including steroidal lactones (known as withanolides) and alkaloids. While withanolides are believed to be responsible for many of ashwagandha's proposed effects, evidence from preclinical studies suggests that other, non-withanolide components may also be involved [6-8].

## Efficacy

### Stress and anxiety

Research suggests that ashwagandha extracts may lower stress, anxiety, and cortisol levels. A taskforce created by the World Federation of Societies of Biological Psychiatry and the Canadian Network for Mood and Anxiety Treatments provisionally recommends specific daily doses of ashwagandha root extract for the treatment of generalized anxiety disorder, but they also note that they cannot provide a stronger recommendation without more data.

Results from several clinical trials suggest that ashwagandha extracts may help reduce stress and anxiety. A 2021 systematic review identified seven studies that investigated the use of ashwagandha to treat stress and anxiety [8]. A total of 491 adults, all from India, with either self-reported high stress and anxiety or a diagnosed anxiety disorder, were randomized to take ashwagandha or placebo for 6 to 8 weeks. Six of the studies used extracts made from ashwagandha root alone (three studies, KSM-66), root and leaf (two studies, Sensoril or Shoden), or unspecified parts (one study), while the seventh study used dried root powder made into granules. The ashwagandha dose varied from 240 to 1,250 mg/day of extract or 12,000 mg/day of whole root granules, which is equivalent to 6,000 mg of root powder. Overall, the studies found that ashwagandha significantly reduced stress and anxiety levels (subjectively measured by validated rating scales), reduced

sleeplessness and fatigue, and reduced serum cortisol levels (a stress hormone) when compared with placebo. In several studies, the benefits appeared to be greater with doses of 500 to 600 mg/day than with lower doses.

Results from most of the studies published after this 2021 review also suggest that ashwagandha has a beneficial impact on perceived stress [9-17]. For example, one clinical trial conducted in Florida included 60 men and women (mean age 34 years) who reported experiencing stress. Participants took capsules that contained 225 mg/day or 400 mg/day of a proprietary ashwagandha root and leaf extract (NooGandha) or placebo for 30 days [9]. Compared with participants in the placebo group, those in both ashwagandha groups reported positive effects on stress, anxiety, depression, and food cravings as measured by validated rating scales. In addition, participants who took the 225-mg dose had lower saliva cortisol levels than those in the placebo group.

At two health centers in India, 130 healthy men and women age 20 to 55 years with self-reported stress were randomized to take a sustained-released ashwagandha root extract (Prolanza) or placebo for 90 days [10]. The extract was standardized to contain 15 mg withanolides per 300-mg capsule, and participants took one capsule daily. Compared with those who received placebo, participants who took ashwagandha extract reported improvements in stress levels and sleep quality as measured by validated rating scales. They also had lower serum cortisol levels. In addition, participants reported improvements in psychological well-being, memory, and focus. Another study in India randomized 54 participants with mild to moderate stress and anxiety to receive either ashwagandha root extract (Shagandha) or placebo [14]. The participants in the ashwagandha group were given tablets that were standardized to contain 2.5% withanolides; each tablet included 500 mg of the root extract and 5 mg of piperine. At day 60, participants in the ashwagandha group had significantly lower scores for stress and anxiety on two validated rating scales than those in the placebo group. In addition, the quality of life scores increased significantly for people in the ashwagandha group between baseline and day 60, and the researchers noted improvements in multitasking and concentration among the participants in this group.

At the University of Colorado, Colorado Springs, 60 students (age 18–50 years) were randomized to take an ashwagandha root extract (Gaia Herbs) or placebo for 30 days in a double-blind trial [11,12]. The extract contained 2.5 mg withanolides per 350-mg capsule, and participants took two capsules daily. The investigators gathered qualitative, subjective information from participants during daily check-ins and focus groups. Participants who took ashwagandha root extract reported increased well-being, including a sense of calm; improved energy levels; heightened mental clarity; and enhanced sleep quality. While the descriptions of stress were comparable in both groups, participants who took ashwagandha were more likely to describe their stress as manageable compared with those taking placebo.

Another randomized clinical trial included 120 healthy men and women (mean age 54–55 years) who were overweight or mildly obese and experiencing low energy and fatigue [17]. Participants took an ashwagandha root extract (Witholytin, which contains 200 mg hydroalcoholic extract of ashwagandha root standardized to 1.5% withanolides) or placebo twice daily for 12 weeks. Compared with placebo, ashwagandha did not reduce perceived stress, but it did reduce fatigue.

A daily dose of 300 to 600 mg ashwagandha root extract (standardized to 5% withanolides) is provisionally recommended for the treatment of generalized anxiety disorder by an international taskforce created by the World Federation of Societies of Biological Psychiatry (WFSBP) and the Canadian Network for Mood and Anxiety Treatments (CANMAT) [18]. The taskforce issued this recommendation in 2022, and they noted that more research is needed to support a stronger recommendation.

## Sleep

Evidence from a small number of studies suggests that taking ashwagandha extract may improve several aspects of sleep, including sleep quality, sleep efficiency, total sleep time, and sleep latency. These benefits tend to be more pronounced among people with insomnia.

Research is limited, but the results from a few clinical trials suggest that ashwagandha extracts may help with sleep. For example, at one study center in India, 150 healthy men and women age 18 to 65 years with self-reported sleep problems characterized by insomnia and lack of restful sleep were randomized to take an ashwagandha root and leaf extract (Shoden) or placebo for 6 weeks [19]. The extract was standardized to

contain 21 mg of withanolide glycosides per 60-mg capsule, and participants took two capsules each day. Both groups reported improvements in sleep quality as measured by a validated rating scale, but the improvements were greater in the ashwagandha group (72%) than in the placebo group (29%). In addition, participants who took ashwagandha extract showed improvements in sleep efficiency (time in bed spent in sleep), total sleep time, sleep latency (time taken to fall asleep), and awakening after sleep onset as assessed by actigraphy, which involves wearing a watch monitor on the wrist to measure body motion. They also reported improvements in quality of life.

In another trial conducted in India, 80 healthy men and women age 18 to 50 years, half of them with insomnia, were randomized to take an ashwagandha root extract (KSM-66) or placebo for 8 weeks [5]. The extract was standardized to a withanolide content of more than 5% per 300-mg capsule, and participants took two capsules each day. Participants with insomnia who took ashwagandha extract showed improvements in sleep quality, sleep onset latency, mental alertness on rising, and perceived anxiety symptoms compared with those taking placebo, as measured by actigraphy and validated rating scales. Participants without insomnia who took ashwagandha also reported that ashwagandha improved their sleep but not their perceived anxiety symptoms or their mental alertness on awakening.

A 2021 systematic review and meta-analysis included five studies (including the two described above) that investigated using ashwagandha to promote sleep [20]. All the studies were conducted in India. A total of 372 adults, either self-described as healthy or with insomnia, took ashwagandha or placebo for 6 to 12 weeks. The dose of the ashwagandha supplement used in these studies ranged from 250 to 600 mg/day as a root extract (KSM-66) or, in one study, 120 mg/day of a root and leaf extract (Shoden). Overall, the studies found that ashwagandha extract had a small but significant effect on improving sleep compared with placebo. The benefits were more prominent when the dose was 600 mg/day and when the treatment duration was at least 8 weeks. Benefits were also more prominent in participants with insomnia.

## Safety

Ashwagandha appears to be well tolerated for up to about 3 months of use, but its long-term safety is not known. Common side effects (e.g., loose stools, nausea, drowsiness) are usually mild. However, there are a few reports of more serious side effects, including adverse effects on liver function. Ashwagandha might also affect thyroid function and interact with some medications. Some experts advise against the use of ashwagandha by women who are pregnant and by men with hormone-sensitive prostate cancer.

In the studies described above and in many other clinical trials, ashwagandha has been well tolerated by participants for up to about 3 months of use. Common side effects are mild and include stomach upset, loose stools, nausea, and drowsiness [8,21]. Increased heart rate variability has also been reported [17]. Evidence on the safety of using ashwagandha over many months or years is lacking.

There are a few reports of more serious side effects associated with ashwagandha use, including adverse effects on liver function [22]. In a 2017 report of liver injury that was associated with ashwagandha use, a 20-year-old man in Japan developed liver dysfunction and hyperbilirubinemia after using ashwagandha (plant part and dose not specified) in combination with multiple antianxiety drugs [23]. Since then, the use of ashwagandha has been linked to acute liver injury in other individuals, some of whom had pre-existing liver disease [24-28]. These include five cases of people (three men and two women, age range 21–62 years) who reportedly took supplements that contained 450 to 1,350 mg ashwagandha (plant part not specified) daily over the course of 1 week to 4 months and experienced signs of liver injury, such as jaundice, pruritus, nausea, lethargy, abdominal discomfort, and hyperbilirubinemia [25]. In these cases and others, the conditions of the individuals improved over time after they stopped taking the supplement; some also received medical treatment [23,25-27]. However, the contents of the products that the individuals took were not independently verified in all cases, and some products were combination products that contained ashwagandha and other ingredients.

Some research suggests that ashwagandha might affect thyroid function. In one study, three adult men who took 500 mg/day of a standardized ashwagandha root and leaf extract for 8 weeks had small increases in blood thyroxine (T4) levels [29]. A small clinical trial with 50 participants with subclinical hypothyroidism found that ashwagandha root extract, at a dose of 300 mg twice daily for 8 weeks, lowered serum thyroid-stimulating hormone (TSH) and increased triiodothyronine (T3) and T4 levels compared with placebo [30].

Three case reports have also described thyrotoxicosis in women who were taking ashwagandha extract (age range 32–73 years) [31]. However, the doses of ashwagandha were not specified in two of these case reports, and the third case involved an unusually high dose (1,950 mg/day for more than 2 months) [32]. Discontinuing ashwagandha resolved the symptoms of thyrotoxicosis in all three cases. These findings suggest that ashwagandha might interact with thyroid hormone medications. Ashwagandha might also interact with other medications, including antidiabetes medications, antihypertensives, immunosuppressants, and sedatives [8,33].

Some experts advise against the use of ashwagandha by women who are pregnant because some reports have suggested that it has the potential to cause spontaneous abortion [8,33-38]. A risk assessment released by the Technical University of Denmark in 2020 discussed both this potential abortifacient effect of ashwagandha and potential effects on thyroid and sex hormones; Denmark subsequently banned the use of ashwagandha in 2023 [39,40]. The French Agency for Food, Environmental, and Occupational Health and Safety (ANSES) issued a statement in 2024 that recommended against using ashwagandha in certain populations, including pregnant and breastfeeding women and people with endocrine disorders [41]. One of the sources for this information is a monograph on ashwagandha root published in 2000 by the American Herbal Pharmacopoeia (AHP) [38]; however, AHP has stated that their report has been misrepresented and that there is no evidence that ashwagandha root causes abortions [42]. Ashwagandha use might also increase testosterone levels [8,17,43,44], so according to experts, it might not be safe for men with hormone-sensitive prostate cancer [36,37].

## Implications for use

Ashwagandha extracts may help reduce anxiety and stress and improve sleep. However, it is challenging to develop recommendations for using ashwagandha, because the studies that have evaluated it have used various preparations and doses. Ashwagandha appears to be well tolerated during short-term use, but the long-term safety of ashwagandha is not known. In addition, ashwagandha may not be safe for certain people.

Several randomized, placebo-controlled clinical trials, most of them fairly small in size and of short duration, have found that ashwagandha may reduce perceived stress and anxiety and improve the quality and duration of sleep [7,8,20,45]. Because studies have used various ashwagandha preparations (with different extraction and standardization processes) and doses, it is difficult to identify specific extracts or recommended amounts [7,46]. In addition, most studies have been conducted as part of a traditional medical system, so the potential effects of ashwagandha when used as a dietary supplement outside of that approach remain unclear.

Ashwagandha appears to be well tolerated for up to 3 months of use. However, the efficacy and safety of long-term ashwagandha use over months or years for stress, anxiety, or sleep is not known. In addition, ashwagandha may have potential adverse effects on the liver and thyroid and might not be safe for men with prostate cancer or women who are pregnant or breastfeeding.

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